

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer shipping full-stack product features end to end (Next.js/TypeScript + Firebase/GCP) on a provider-facing **SMART on FHIR** clinical decision support app that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, exposing them through API endpoints while **Gemini** is constrained to synthesis and the backend deterministically controls eligibility and coverage.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus **REST** integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack internal annotation platform (React, Node.js, MongoDB) used daily by the AI team, replacing spreadsheet workflows with secure role-based REST APIs and cutting manual operations by **100+ hours/month** while lifting labeling throughput **60%**.
- Chased production issues across tenants from log and error signals to shipped fixes, monitoring releases and client integrations and resolving tenant-specific incidents to keep the multi-tenant eTriage service stable.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with **Stripe** + manual-cash hybrid payments and a partial-payment ledger, plus cron-based background jobs for renewal automation; live in production **3+ years** with git-webhook auto-deploy.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that turns open-ended queries into structured, provenance-tracked entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback; hardened with **150+ automated tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI assistant end to end in **TypeScript** on Cloudflare Workers, streaming responses to the client over WebSockets in real time and running durable multi-step **background workflows** that survive process restarts and tool failures via Durable Objects and D1.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark **data-pipeline** runtime **~25%** (from 17–19 min to 14.3 min) on a **~1.4B-row** NYC TLC dataset via broadcast joins, AQE, skew-join handling, and partition/column pruning, using Spark UI stage timings to localize shuffle and skew bottlenecks.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: TypeScript, JavaScript, Python, SQL, C++, Java, Bash

Full-Stack: Next.js, React, Redux, Tailwind CSS, Vite, HTML/CSS

Backend & APIs: Node.js, Express, FastAPI, REST APIs, Stripe, async services / background jobs

Databases: PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.